College Physics 2

Lights and Optics

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College Physics 2

Properties of Light

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OBJECTIVES

- Explain the nature of light

The Electromagnetic Spectrum

Like all waves, electromagnetic waves have a wavelength and frequency, related by:

$$c = f\lambda$$

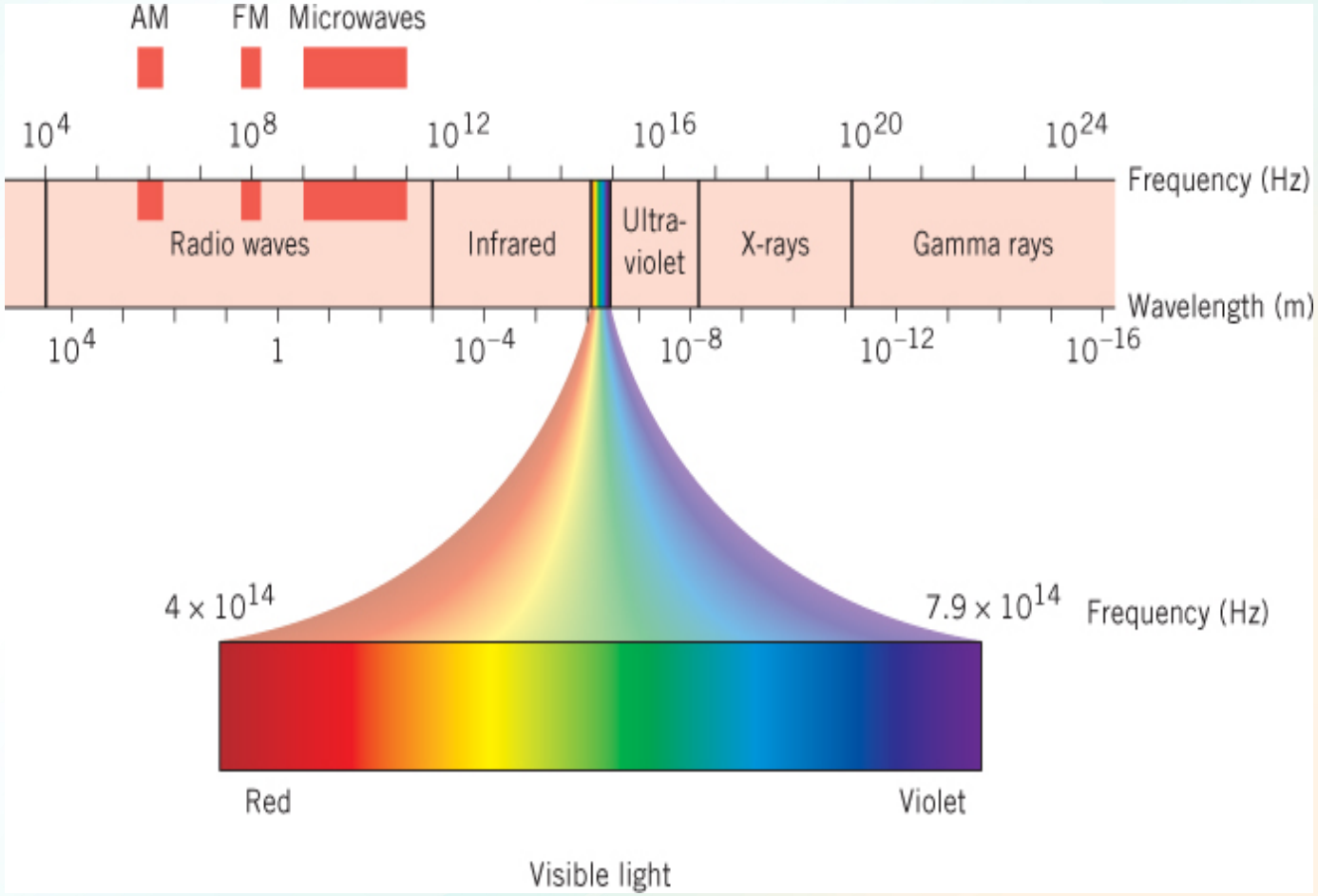


Figure 1. Electromagnetic spectrum. From. *http://hep0.okstate.edu/flera/phys1214/Ch22_Giancoli7e.pdf*

Speed of Light

The speed at which all light (all parts of the electromagnetic spectrum) travels in a vacuum.

$c = 2.99792458 \times 10^8 m/s$

The duration of one second is defined by the cesium clock, so one meter is defined to be the distance that light travels in 1/299,792,458 s.

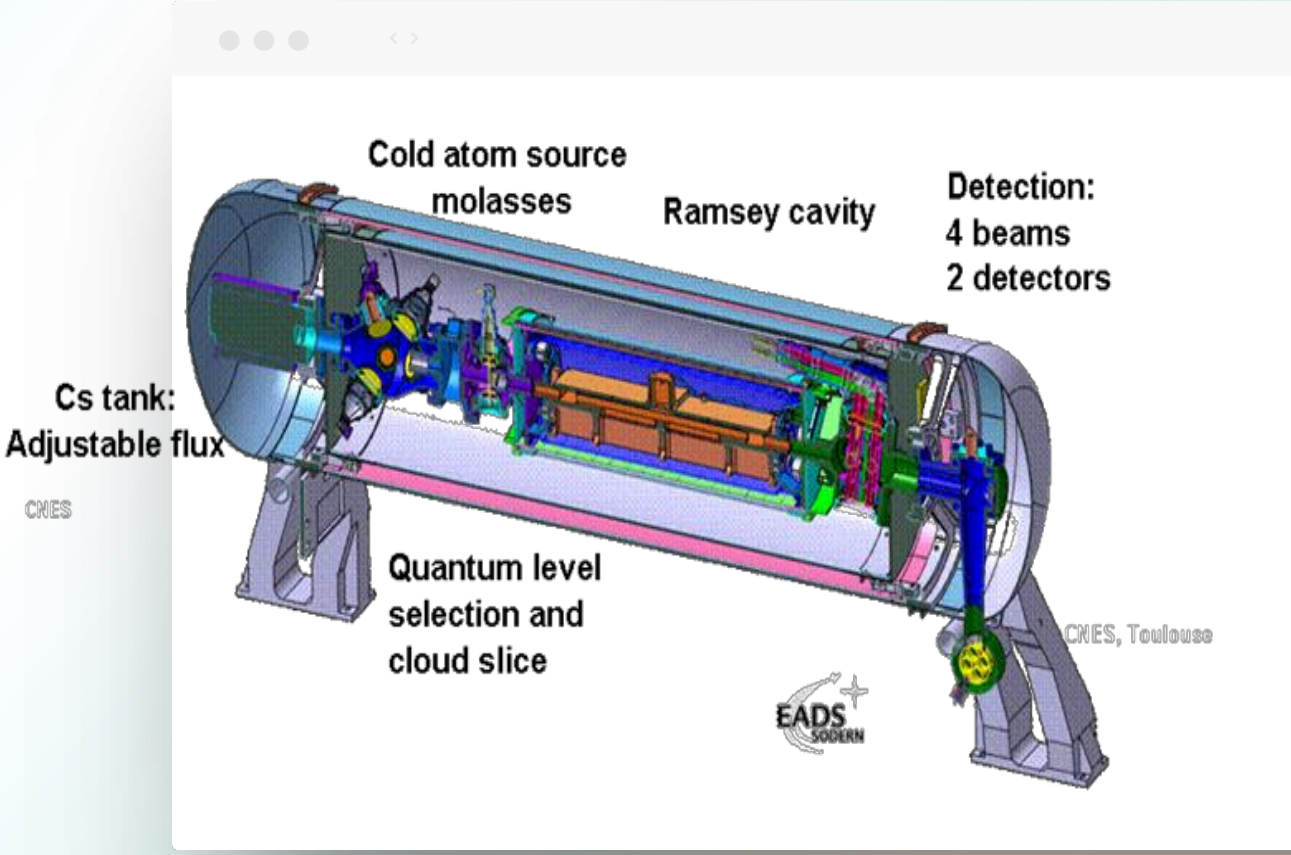


Figure 2. Cesium clock. From. *https://en.wikipedia.org/wiki/Sodern*

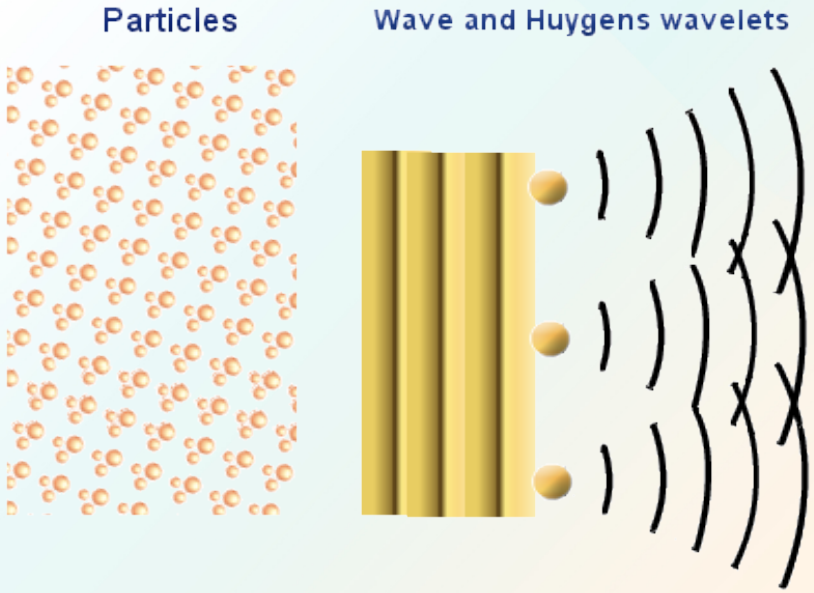
NATURE OF LIGHT

Before 19th century:

Light was considered to be a stream of particles. The particles were either emitted by the object being viewed or emanated from the eyes of the viewer. Newton was the chief architect of the **particle theory of light**. He believed the particles left the object and stimulated the sense of sight upon entering the eyes.

During the 19th century:

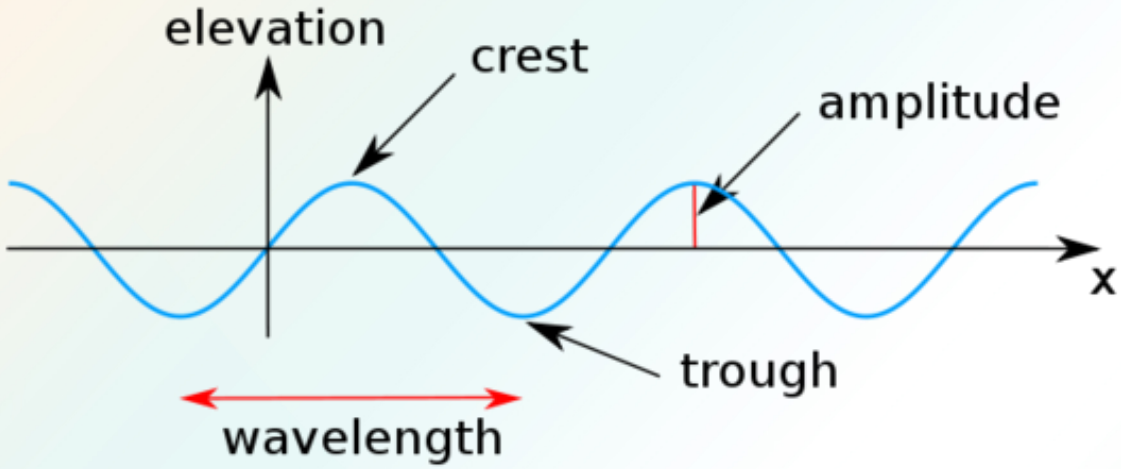
Other developments led to the general acceptance of the **wave theory of light.** Christian Huygens argued that light might be some sort of a wave motion. Thomas Young (in 1801) provided the first clear demonstration of the wave nature of light. He showed that light rays interfere with each other. Such behavior could not be explained by particles.



Dual nature of Light

Figure 3. Light as particles and wave. From. *https://tse4.mm.bing.net/th?id=OIP.b1Zx6s1iOWsZoDZQhD0kwgHaGM&pid=Api&P=0&w=202&h=170*

Wavelength



- Propagation of light is best described by a wave model.
- Can be expressed in any unit of length, usually in meters (m) and nanometer (nm).
- Represented by lambda (λ).

Figure 4. Wavelength of light. From. h*ttps://chem.libretexts.org/@api/deki/files/8879/waves.png?revision=1&size=bestfit&width=342&height=200*

Sample Problem

Find the range in wavelengths for visible light in the frequency range between 4.0×10^{14}Hz and 7.9×10^{14}Hz

Solution:

$$\lambda = \frac{c}{f} = \frac{3.00 \times 10^{8}\ \text{m/s}}{4.0 \times 10^{14}\ \text{Hz}} = 7.5 \times 10^{-7}\ \text{m} = 750\ \text{nm}$$

$$\lambda = \frac{c}{f} = \frac{3.00 \times 10^{8}\ \text{m/s}}{7.9 \times 10^{14}\ \text{Hz}} = 3.8 \times 10^{-7}\ \text{m} = 380\ \text{nm}$$

NATURE OF LIGHT

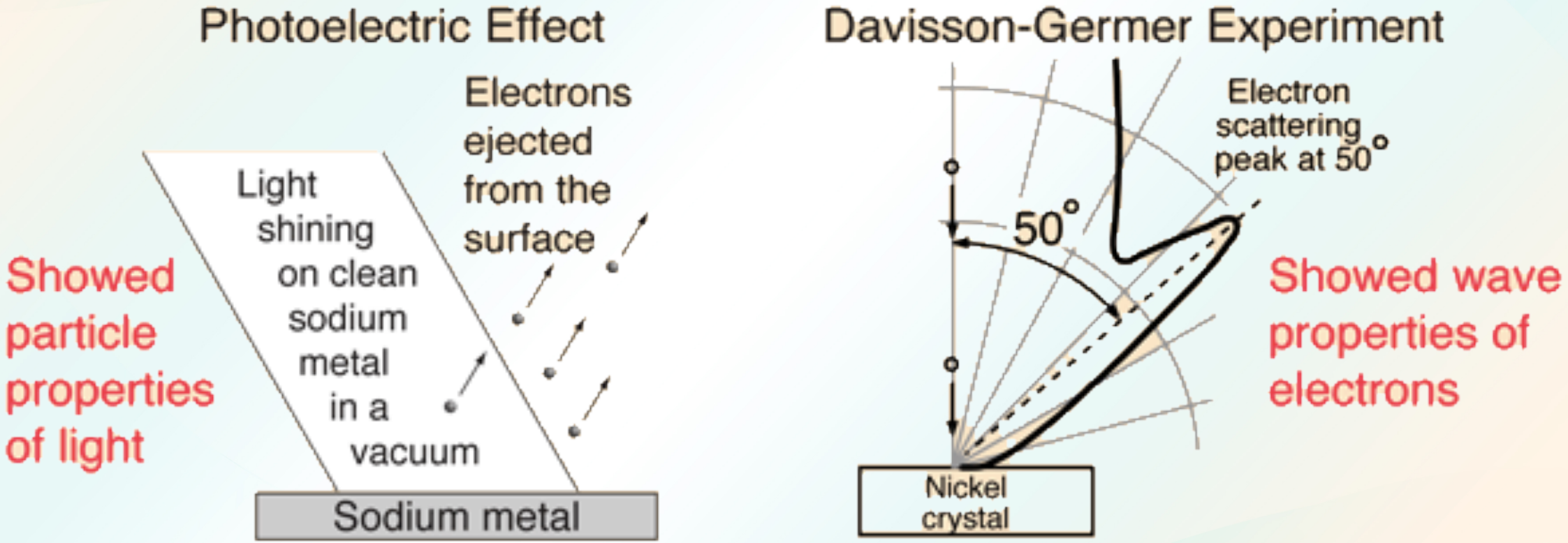


Figure 4. Nature of light. From. *https://tse1.mm.bing.net/th?id=OIP.JhBqInK-GvH66ys5KQKItQHaCs&pid=Api&P=0&w=479&h=175*

NATURE OF LIGHT

The photoelectric effect was a major phenomenon not explained by waves.
When light strikes a metal surface, electrons are sometimes ejected from the surface.
The kinetic energy of the ejected electron is independent of the frequency of the light.
Einstein (in 1905) proposed an explanation of the photoelectric effect that used the idea of quantization.

The quantization model assumes that the energy of a light wave is present in particles called photons.

$E = hf$ where h is Planck's Constant and = 6.63×10^{-34} J·s

In view of these developments, light must be regarded as having a dual nature.
Light exhibits the characteristics of a wave in some situations and the characteristics of a particle in other situations.

For practical applications of geometric optics, light is treated as a wave phenomenon.

NATURE OF LIGHT

Ray optics (sometimes called *geometric* optics) involves the study of the propagation of light.

It uses the assumption that light travels in a straight-line path in a uniform medium and changes its direction when it meets the surface of a different medium or if the optical properties of the medium are non-uniform.

The ray approximation is used to represent beams of light.

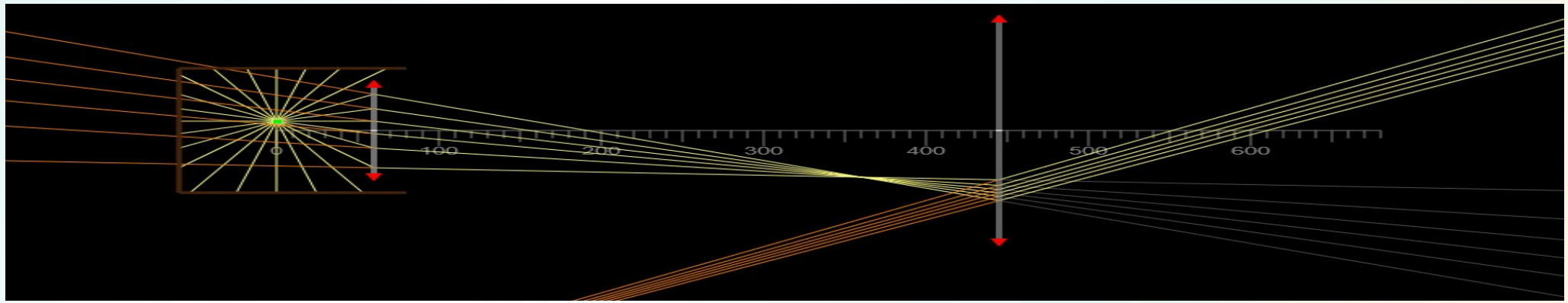


Figure 5. Ray optics. From. *https://tse1.mm.bing.net/th?id=OIP.MqTe-1fSg7VRhVO_DlHuJAHaDe&pid=Api&P=0&w=436&h=205*

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College Physics 2

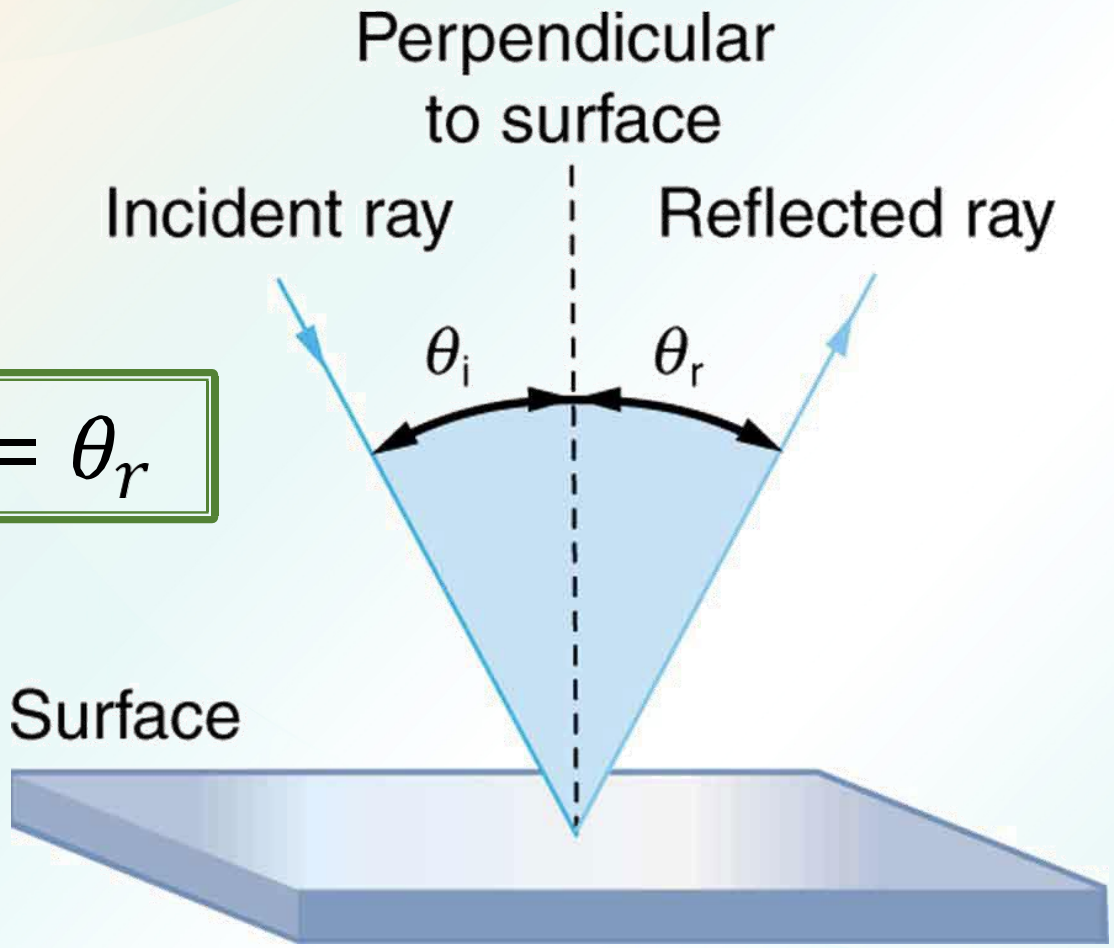
Reflection, Refraction, Snell's Law & Total Internal Reflection

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OBJECTIVES

- Discuss the Law of Reflection and Law of Refraction
- Explain Snell's Law
- Use Snell's law in determining angle of refraction
- Explain Total Internal Reflection

Law of Reflection



$$\theta_i = \theta_r$$

> **The angle of reflection is equal to the angle of incidence.**

Figure 6. Law of reflection. From. *http://cnx.org/resources/55c2011fd4cdc676168cc52e7aadaff7/Figure%2026_02_01.jpg*

Law of Reflection

The reflection of light can be roughly categorized into two types of reflection:

Specular reflection --- light reflected from a smooth surface at a definite angle.

Diffuse reflection --- produced by rough surfaces that tend to reflect light in all directions.

Refraction

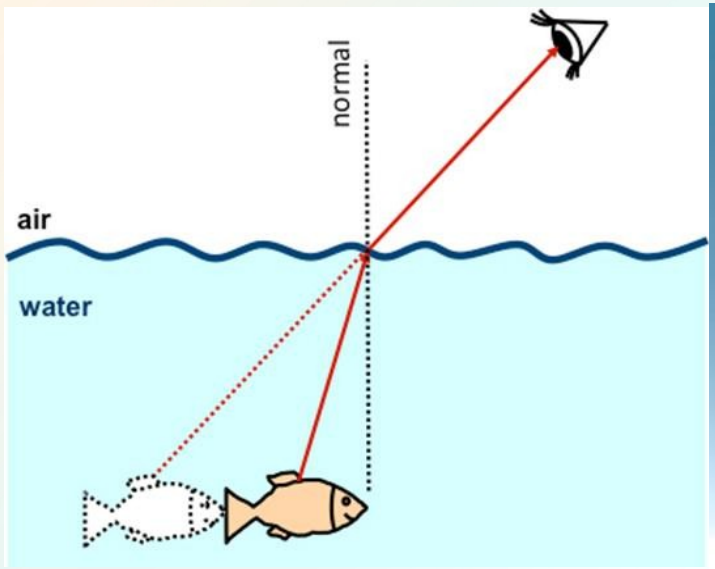




Figure 7. Refraction. From. *https://wiki.anton-paar.com/fileadmin/wiki/images/basics-of-refractometry/refraction-of-light.jpg*

Refraction is the bending of the ray of light as it passes the boundary of two media.

Refraction is caused by the change in speed experienced by a wave when it changes medium.

INDEX OF REFRACTION

The speed of light in a vacuum is:

$$c = 3.00\ x\ 10^8 m/s$$

In other transparent materials, speed (*v*) is always less than that of the vacuum.

Index of refraction, n, varies with medium.

$$n = \frac{c}{v}$$

> The ratio of the speed of light in vacuum to the speed in a given material is called the index of refraction.

INDEX OF REFRACTION

Material medium	Refractive index	Material medium	Refractive index
Air	1.0003	Fused quartz	1.46
Canada Balsam	1.53	Turpentine oil	1.47
Ice	1.31	Benzene	1.50
Water	1.33	Crown glass	1.52
Rock salt	1.54	Diamond	2.42
Alcohol	1.36	Dense flint glass	1.65
Kerosene	1.44	Ruby	1.71
Carbon disulphide	1.63	Sapphire	1.77

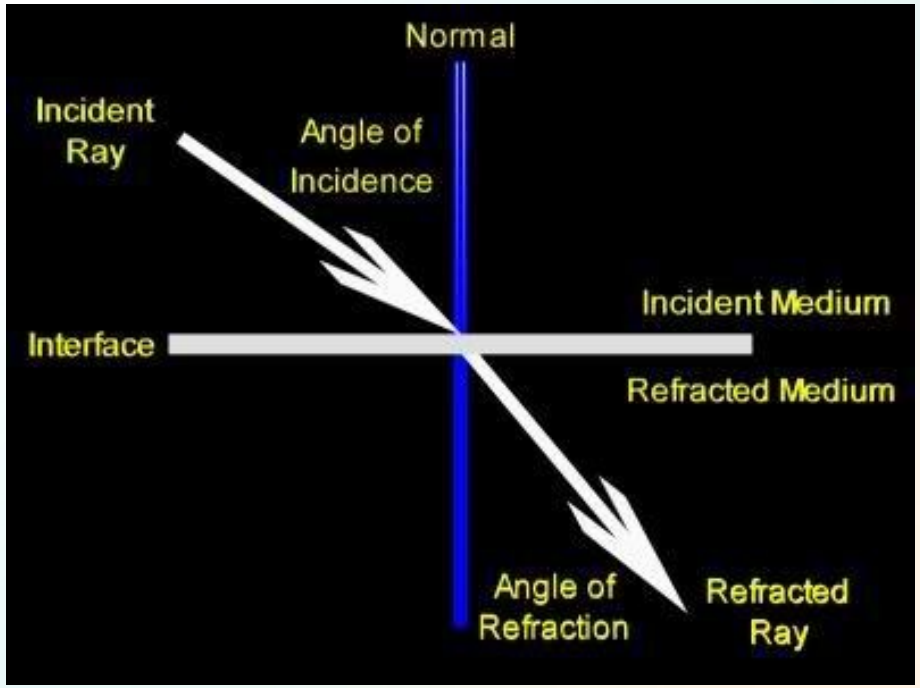
LAWS OF REFRACTION

The incident ray, the refracted ray and the normal to the surface separating the two media at the point of incidence, all lies on the same plane.

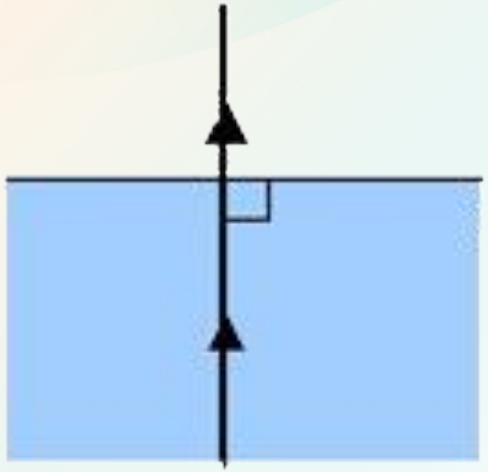
When light travels from a medium n_1 into another medium n_2,

- when $n_2 > n_1$, light will refract toward the normal.

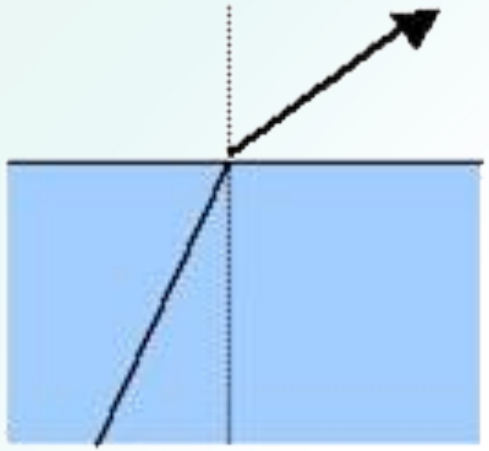
- when $n_1 > n_2$, light will refract away from the normal.



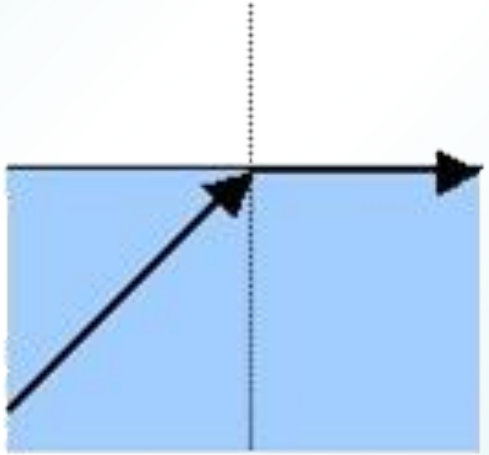
LAWS OF REFRACTION



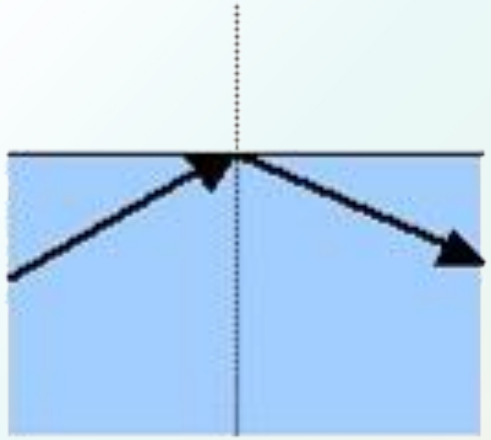
Angle of incidence is 0^0 from normal, Light enters the next medium at 0^0 from from normal also.



Angle of incidence less than the critical angle. Refraction occurs.



Angle of incidence is equal to the critical angle. Light is refracted along the boundary.



Angle of incidence greater than the critical angle. Total internal reflection occurs.

SNELL'S LAW

The angle of refraction depends on the speed of light in the two media and on the incident angle.

Snell's Law is the basic of law of refraction.

$$n_1 sin\theta_1 = n_2 sin\theta_2$$

Where n_1 = index of refraction for medium 1
n_2 = index of refraction for medium 2
θ_1 = angle of incidence
θ_2 = angle of refraction

> **A ray of light bends in such a way that the ratio of the sine of the angle of incidence to the sine of the angle of refraction is constant.**

SNELL'S LAW

If light enters a medium where n is greater (its speed is less), then the ray is bent toward the normal.

$$n_2 > n_1 \rightarrow \theta_2 < \theta_1$$

If light enters a medium where n is lesser (its speed is higher), then the ray is AWAY from the normal.

$$n_2 < n_1 \rightarrow \theta_2 > \theta_1$$

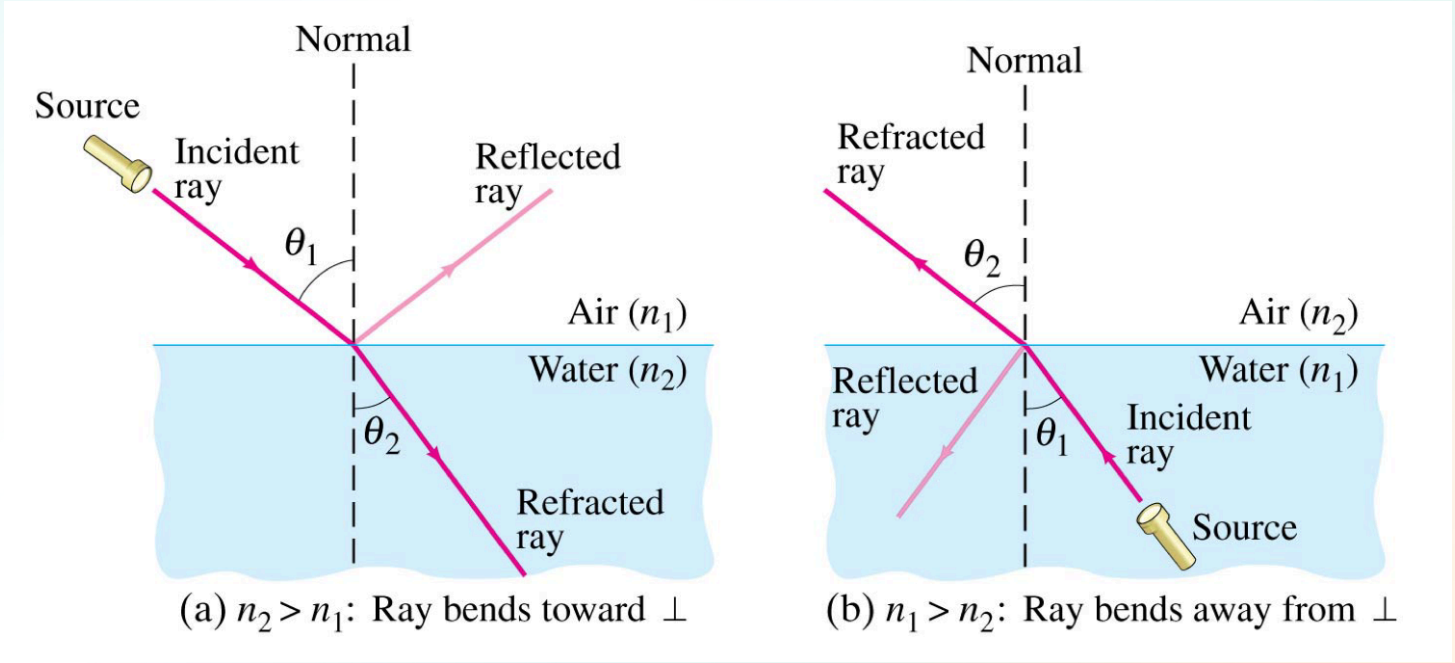


Figure 8. Snell's Law. From. *http://hep0.okstate.edu/flera/phys1214/Ch22_Giancoli7e.pdf*

Sample Problems

1. Light traveling in air strikes the water at an incident angle of 45°. What is the angle of refraction?

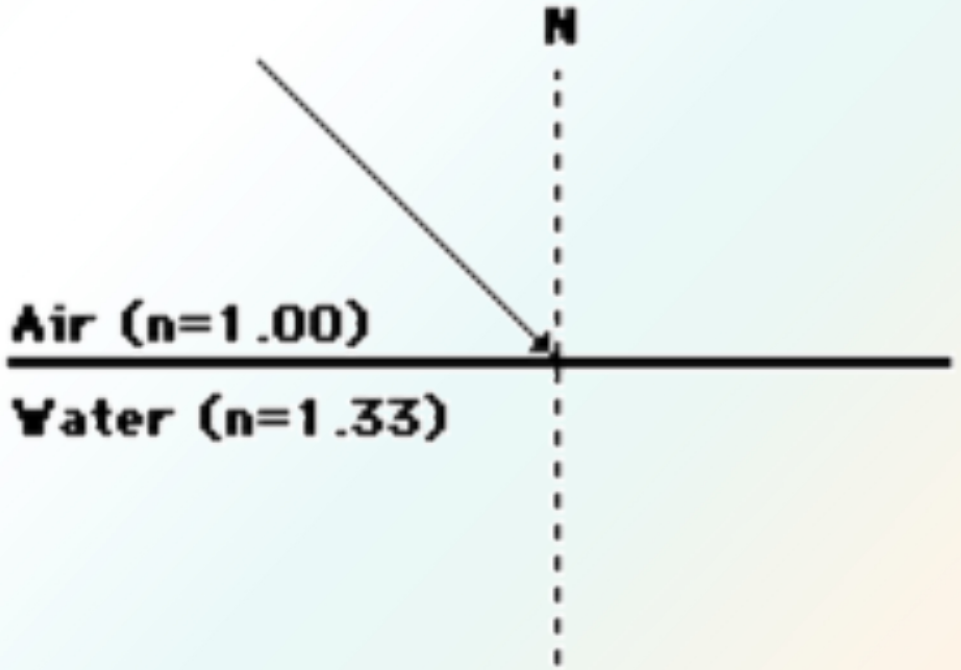
$$n_1 sin\theta_1 = n_2 sin\theta_2$$

$$\sin\theta_2 = \frac{n_1 sin\theta_1}{n_2}$$

$$\theta_2 = \sin^{-}\left(\frac{n_1 sin\theta_1}{n_2}\right)$$

$$\theta_2 = \sin^{-}\left(\frac{(1)\sin 45}{1.33}\right)$$

$$\boldsymbol{\theta_2 = 32^0}$$



Sample Problems

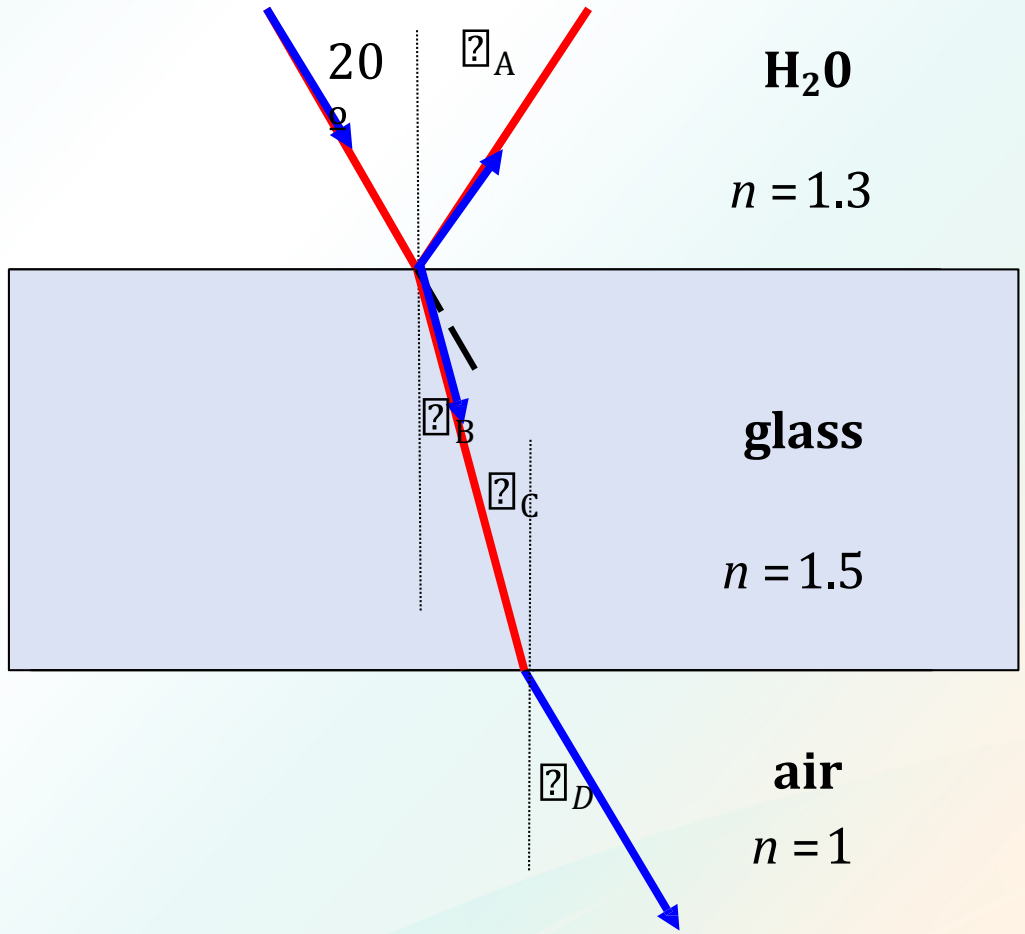
2. Light passes through water then to a thick glass and to air as illustrated.

A. Determine the angle of the reflected light in water, θ_A.

B. Determine the angle of refraction of light in the glass, θ_B.

C. Determine the angle of incidence of light as it exits the glass, θ_C.

D. Determine the angle of refraction of light as it enters the air, θ_D.



Solution

A. $\theta_A = 20^0$

B. $\theta_B =$ 17.24^0

$$n_i sin\theta_i = n_B sin\theta_B$$

$$\theta_B = \sin^{-}\left(\frac{n_i sin\theta_i}{n_B}\right)$$

$$\theta_B = \sin^{-}\left(\frac{(1.3)(\sin 20)}{1.5}\right)$$

$$\boldsymbol{\theta_B = 17.24^0}$$

A. $\theta_C =$ 17.24^0

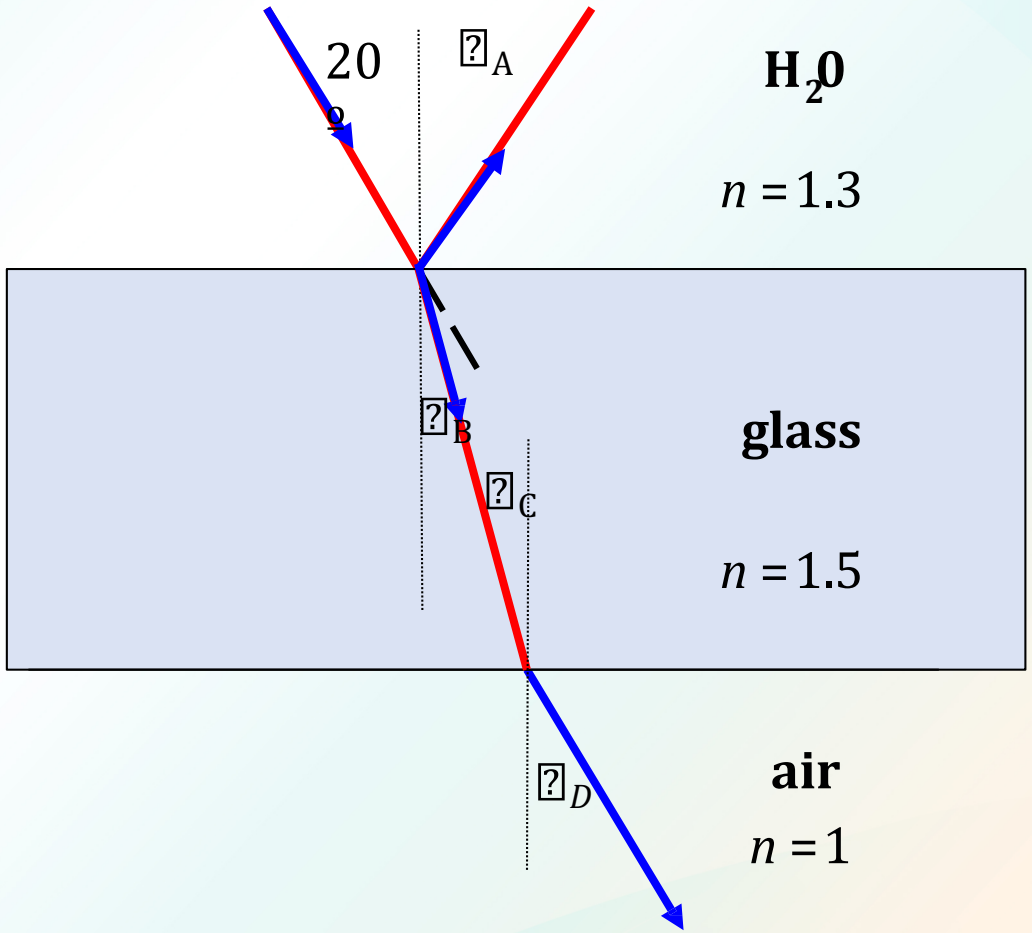
B. $\theta_D = 20^0$

$$n_C sin\theta_C = n_D sin\theta_D$$

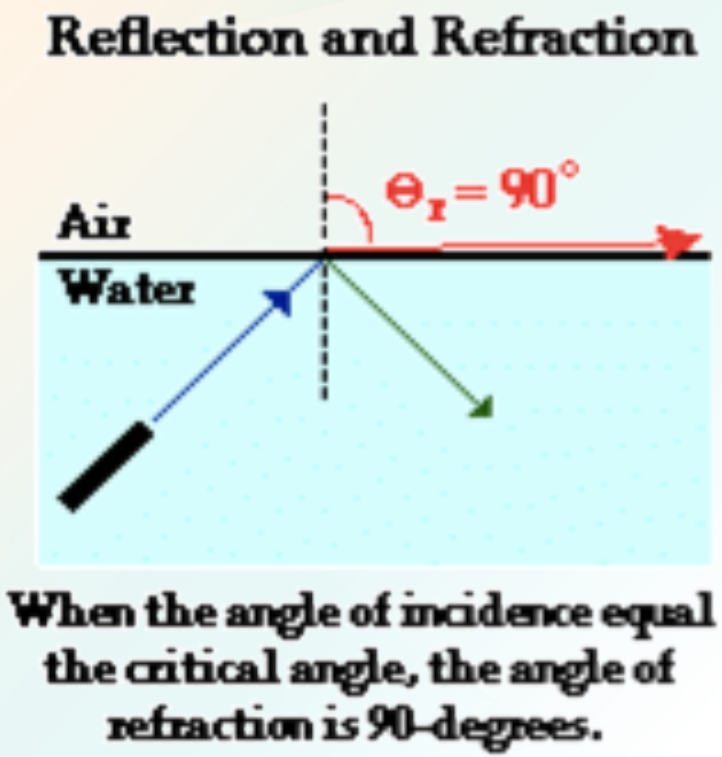
$$\theta_D = \sin^{-}\left(\frac{n_C sin\theta_C}{n_D}\right)$$

$$\theta_D = \sin^{-}\left(\frac{(1.5)(\sin 17.24)}{1}\right)$$

$$\boldsymbol{\theta_D = 26.40^0}$$



CRITICAL ANGLE



$$n_1 sin\theta_1 = n_2 sin\theta_2$$

$$\theta_1 = sin^{-1}\left(\frac{n_2 sin\theta_2}{n_1}\right)$$

$$\theta_1 = \theta_C$$

$$\theta_2 = 90^0$$

$$\theta_c = sin^{-1}\left(\frac{n_2 sin90}{n_1}\right)$$

$$\boldsymbol{\theta_c = sin^{-1}\left(\frac{n_2}{n_1}\right)}$$

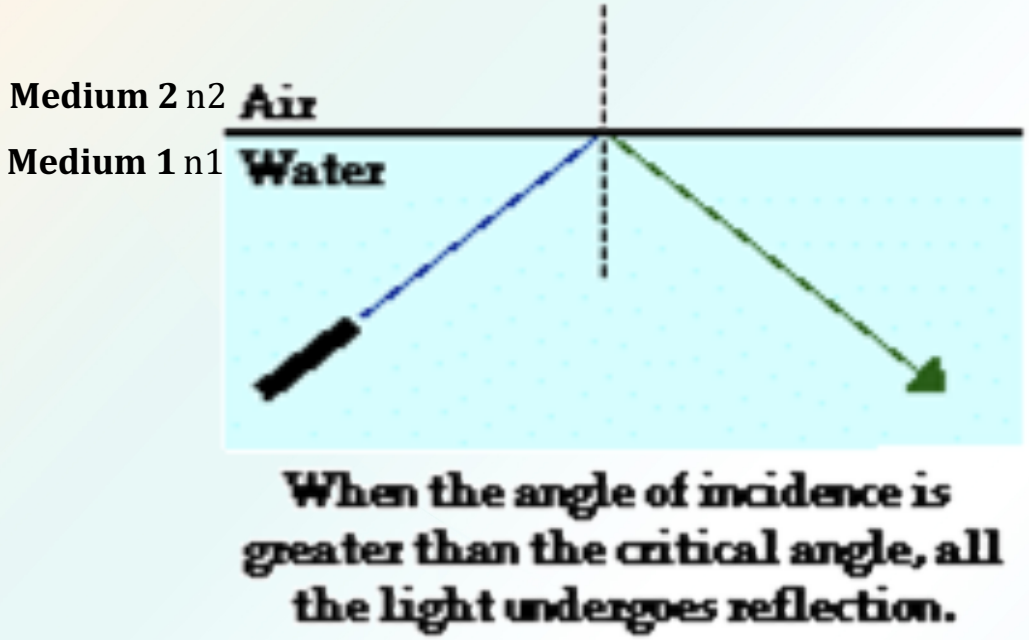
The critical angle is the angle of incidence that produces an angle of refraction of 90º

Figure 9. Reflection and refraction. From. *https://habibs.files.wordpress.com/2008/03/critical_angle.gif?w=544*

TOTAL INTERNAL REFLECTION

$$\theta_i > \theta_c$$

$$n_1 > n_2$$



No refraction, only reflection.

Total internal reflection occurs when light attempts to pass from a more optically dense medium to a less optically dense medium at an angle greater than the critical angle.

Figure 10. Total Internal reflection. From. *https://habibs.files.wordpress.com/2008/03/critical_angle.gif?w=544*

TOTAL INTERNAL REFLECTION

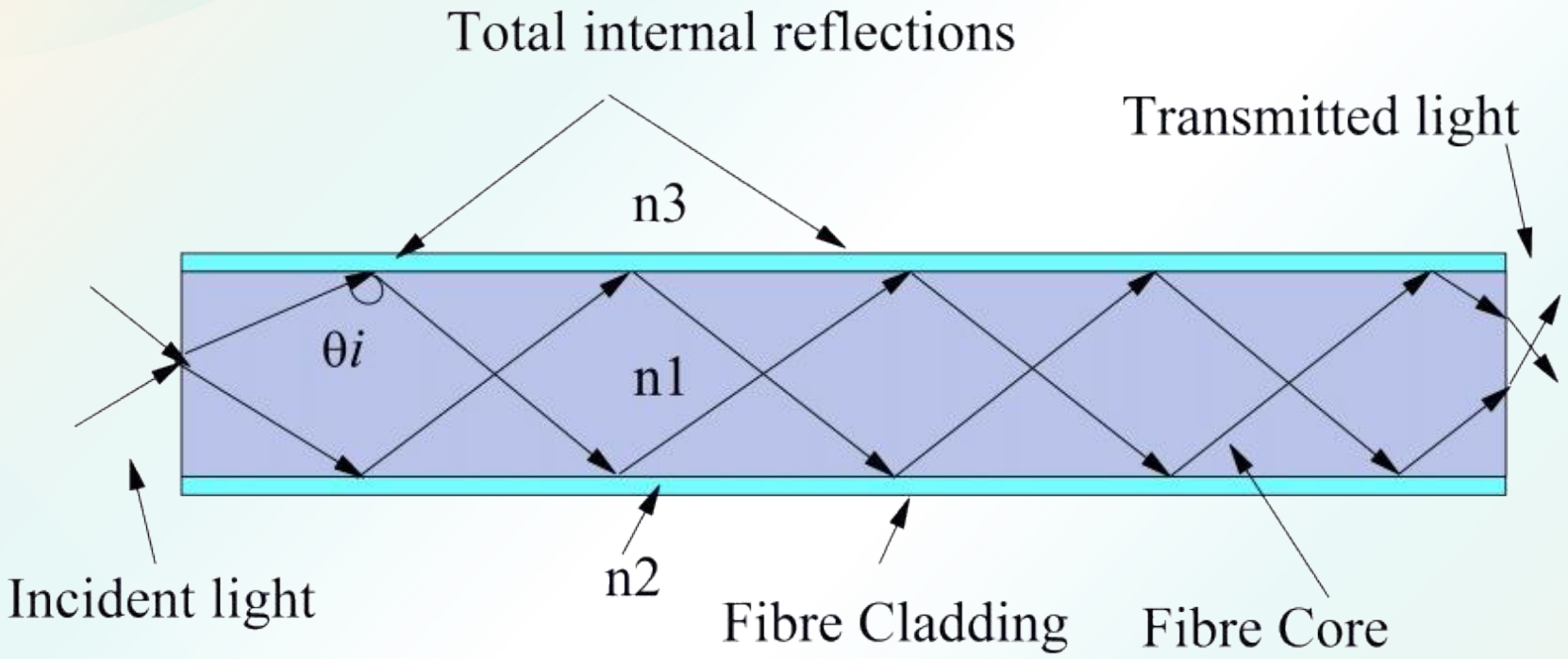


Figure 11. Fiber optic total internal reflections. From. *https://tse1.mm.bing.net/th?id=OIP.Jkc_QRWDu0Gd4GN5espOfAHaDL&pid=Api&P=0&w=380&h=164*

Cladding has lower refractive index.
Core has higher refractive index.

TOTAL INTERNAL REFLECTION

Light travels through the core of a fiber optic by continually reflecting off of the cladding.

Due to total internal reflection, the cladding does not absorb any of the light, allowing the light to travel over great distances.

Fiber optic lines are strands of glass or transparent fibers that allows the transmission of light and digital information over long distances. They are used for the telephone system, the cable TV system, the internet, medical imaging, and mechanical engineering inspection.

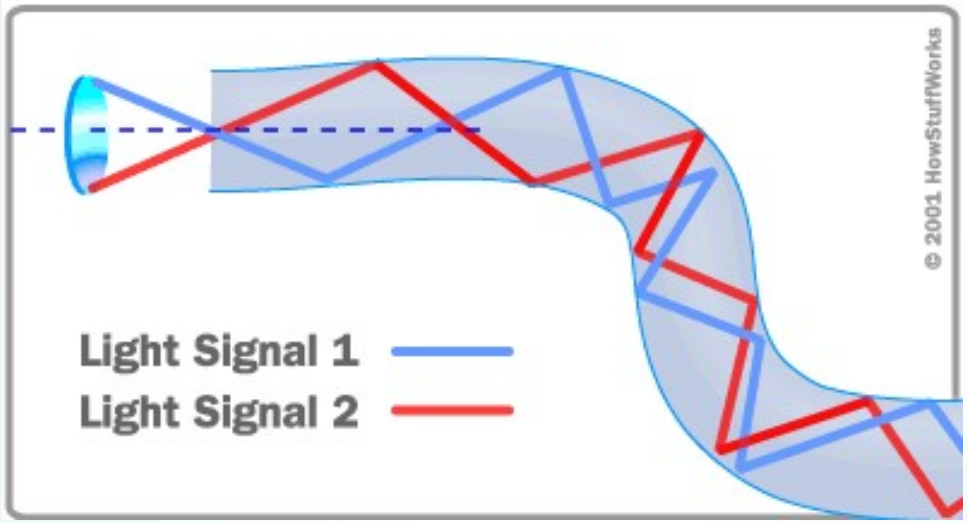


Figure 12. Fiber optic lines with two light signals. From *https://tse4.mm.bing.net/th?id=OIP.QeU4_lp3S2PU1mNLmh3CtgAAAA&pid=Api&P=0&w=382&h=207*

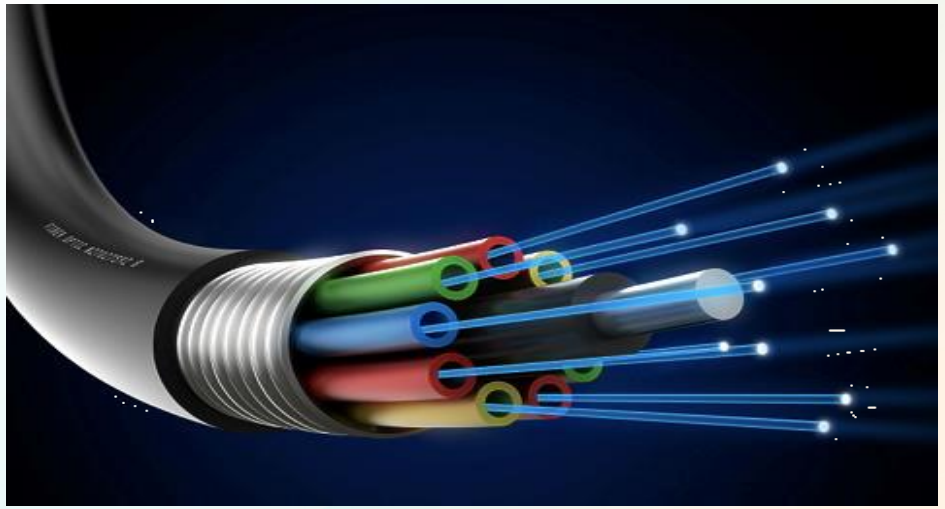


Figure 13. Fiber cable. From. *https://tse1.mm.bing.net/th?id=OIP.NW9Km9vq-toUw5w5Q6OfsQHaEK&pid=Api&P=0&w=305&h=172*

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